HL-CYB-001

T1 report - Project bootstrap and GitHub repository setup

Enterprise Zero Trust Security Architecture

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1. A. Task Information

Table 1. A. Task Information

Field	Record
Task ID	T1 (Task 001)
Title	Project bootstrap and GitHub repository setup
Execution date	2026-11-05, America/Toronto
Scope and objectives	Create the planning/documentation scaffold, establish safe public GitHub publication, and verify repository checks and synchronization
Dependencies	Owner Task 001 and GitHub addendum; available Git/Python/GitHub CLI/Git Credential Manager; authenticated repository owner
Evidence cutoff	Original report at `4186787b153b2967381cffab353aaa81c72e6f70`, retained source appendix, bootstrap evidence, and verified bootstrap CI metadata
Migration	Reporting-governance addendum, 2026-11-05; source preserved without changing historical acceptance or test outcomes
Finalized on	2026-11-05, America/Toronto; historical evidence reviewed and preservation checks passed

The original project report source is authoritative. The source appendix preserves every byte of that UTF-8 Git blob. This structured account reorganizes its information and links to the existing evidence; it does not create operational results.

2. B. Implementation Summary

Task 001 created the repository's documentation and validation framework. It did not inspect hardware, install software, change networks, create VMs, deploy identity/PKI, enroll remote access, or run security tests.

The initial bootstrap created twenty files; its completion-evidence commit added one more. The twenty-one files were `README.md`, `AGENTS.md`, `.gitignore`, `.gitattributes`, `.github/workflows/documentation.yml`, the ten documents in `docs/` (assignment, implementation guide, architecture, security requirements, test plan, acceptance criteria, decisions, risks, final report, references), the infrastructure/configuration/evidence READMEs, `scripts/validate_docs.py`, `tests/test_validate_docs.py`, and `evidence/BOOTSTRAP-VALIDATION.md`.

Repository configuration used public visibility, `main`, the authorized HTTPS origin, repository-local noreply author identity, read-only workflow permissions, a commit-pinned checkout action, and no persistent checkout credentials. No global Git configuration, production setting, software license, or lab configuration was changed. The design remained virtual-first with an offline baseline, symbolic addresses, separate lab identities, and per-stage owner approval.

The reporting migration is documented separately in T1 addendum 001; it is not a claim that those later changes were part of the original bootstrap.

3. C. Testing and Validation

Table 2. C. Testing and Validation

Check / executed command	Expected result	Actual historical result	Status
`python scripts/validate_docs.py`	Required structure, valid local Markdown links/anchors, complete stages/traceability/rubric, workflow and publication checks	Initial twenty-file set passed; final twenty-one-file set passed in bootstrap completion evidence	PASS
`python -m unittest discover -s tests -v`	Focused validator rejection/resolution cases succeed	Twelve tests passed with normal filesystem permissions and on the hosted runner	PASS
`git diff --cached --check`, `git diff --check`	No whitespace errors	Passed; staged text matched reviewed content	PASS
Independent stage/requirement/capacity counts	Consistent planned totals	Sixteen requirements, twelve stages, 36 GiB guest RAM and 248 GiB virtual disks; calculations only	PASS
Public reference URL reachability	Every external reference independently reachable	Thirty-seven of fifty retained URLs returned HTTP success; thirteen could not be verified by the local client because of transport/certificate issues	NOT TESTED for those thirteen; limitation retained
Git/remote/CI read-back	Correct public owner/repository/branch and matching clean local/remote commits	Verified bootstrap publication and CI; details below and in the retained evidence	PASS
Hardware, isolation, firewall, directory, PKI, remote access, logging, and recovery	Approved future implementation and operational evidence	Not performed in Task 001	NOT TESTED

Executed preflight/publication commands included Git status/version/config inspection, `gh auth status`, `gh api user`, repository lookups/creation, safe Git initialization/commit/push, `git remote -v`, `git status`, `git branch -vv`, `git log -1`, `git ls-remote origin`, and workflow metadata reads. Exact observed synchronization commands and outputs are summarized in the unchanged bootstrap evidence. No token is reproduced here.

The initial restricted Windows test run could not access Python-created temporary fixture directories. Tests were rerun successfully with normal filesystem permissions; the hosted Linux test suite also passed. No TLS validation was bypassed to turn unreachable references into passing results.

4. D. Security and Quality Review

Focused scans and authored/staged-content review found no selected publication findings in the reviewed text, commit history, or commit metadata. Actual addresses, credentials, private inventories, raw evidence, production exports, and proprietary source were excluded. The checks are pattern-based and do not prove the absence of every possible secret.

The CLI token's workflow-file push was rejected for missing workflow scope. The existing supported Git Credential Manager login published the same reviewed commit successfully. No token was stored in the

project, no new credential was requested, and no history was overwritten or force-pushed.

Isolation verification: **NOT TESTED** operationally. The offline architecture, protected-network boundary, and stage approval controls were reviewed as documentation only. Hardware, isolated egress, storage/backups, licensing, provider plans, PKI custody, and repository protection remained decisions/risks rather than remediated operational findings.

5. E. GitHub Operations

Table 3. E. GitHub Operations

Item	Historical record
Repository / account	waqashamdani-ca/enterprise-zero-trust-security; `waqashamdani-ca`
Branch / visibility	`main`; PUBLIC
Bootstrap commit	2640261ad2738f1e6d4497ee68636621ed403ea3
Completion-record commit	4186787b153b2967381cffab353aaa81c72e6f70
Pull request	Not applicable: authorized initial bootstrap used direct commits to the new empty repository
CI	Initial run 42139263675: SUCCESS; completion run 42139654765: SUCCESS, observed during bootstrap completion
Merge status	No PR merge; two initial bootstrap commits published without rewriting history
Synchronization	Bootstrap completion read-back found a clean worktree, local `main` tracking `origin/main`, zero ahead/behind, and local/remote HEAD at the completion-record SHA
Permissions / license	Owner administrator/push permissions verified; no license selected

The original report explicitly deferred the completion-record SHA/CI read-back to the original final response. That observed completion metadata is identified here as additional existing bootstrap evidence, not silently inserted into the verbatim source appendix.

6. F. Outstanding Issues

No Task 001 publication blocker remained. Thirteen public reference URLs were not verified by the local HTTPS client. The scanner was deliberately limited, hardware and capacity were not inspected, all operational tests were unexecuted, and no independent human sign-off or owner UAT decision was available.

Open decisions included isolated egress, exact supported versions/evaluation terms, actual private addressing, lab-only provider identities/plans, log transport/retention, CA key custody, backup device/capacity and representative recovery scope, licensing, and repository protection/reviewer settings. See the original source appendix for the historical list; current decisions and risks may evolve without changing this report.

7. G. Completion Status

Table 4. G. Completion Status

Area	Status at the historical evidence cutoff
Engineering	Task 001 documentation/repository bootstrap completed and published
Independent verification	Separate counts, Git remote/API read-back, and hosted CI supported engineering verification; no independent human reviewer sign-off claimed
Owner UAT	PENDING; no explicit owner acceptance supplied
Infrastructure	PLANNED / NOT TESTED; stages A-L not started
Remaining owner decisions	Stage execution approvals, hardware/backup/egress choices, licensing, and repository protection/reviewer model

8. H. Next Task

The original recommendation was one separately approved, read-only Stage A hardware and isolation readiness assessment. It required owner review of Task 001, a bounded approval, device ownership/access, and private evidence handling. This historical recommendation did not authorize Stage A or Task 2. Later owner-authorized documentation work is tracked in its own record.

9. Source preservation appendix

Source: `docs/FINAL-REPORT.md` at `4186787b153b2967381cffab353aaa81c72e6f70`. Source SHA-256: `e53bf1111c5f66bcc4280246792142f82024f77d7d1e5ba4c72e1b06ac67b999`. Links inside the following verbatim source retain their historical relative paths; use the active links above for current navigation.

```
# Task 001 report and future implementation report

## Current project status

| Area | State |
| --- | --- |
| Project | HL-CYB-001 - Enterprise Zero Trust Security Architecture |
| Task 001 | Bootstrap published; documentation checks and initial CI passed; synchronized
publication snapshot recorded below |
| Operational implementation | **NOT STARTED** |
| Stages A-L | **NOT STARTED; separate owner approval required** |
| Owner review / UAT | **PENDING** |
| License | **NOT SELECTED; owner decision required** |

## Bootstrap scope and preflight

The owner authorized planning, scaffolding, documentation checks, safe local Git initialization,
and reviewed public GitHub creation/synchronization. The GitHub addendum fixes the account as
`waqashamdani-ca`, the repository as `enterprise-zero-trust-security`, public visibility, and the
new default branch as `main`.

Preflight found an empty project directory with no Git repository, existing files, ancestor
AGENTS.md instructions, or content conflicts. Git, GitHub CLI, Python, and Node were available.
The initial sandboxed CLI authentication check could not validate the saved login; the
network-enabled check confirmed the existing authenticated `waqashamdani-ca` account without
reauthentication or storing a project token. Both the GitHub integration lookup and authenticated
CLI lookup found no accessible target repository. The CLI account's authenticated identity was
verified before repository creation.

Hardware descriptions remain supplied information, not inspected inventory. No
host/VM/network/security test or implementation readiness assessment was run. No software was
installed, no production system was accessed for testing, and no infrastructure configuration was
changed.

## Created artifact groups

- Root README, governance, public-safe ignore rules, line-ending policy, and documentation-only
workflow.
- Assignment with sixteen measurable requirements, business case, scope, deliverables, standards
mapping references, and a 100-point rubric.
- Proposed architecture and detailed A-L blueprint, including dependencies, expected outcomes,
tests, rollback, and per-stage approvals.
- Security/threat model, functional/negative/security/recovery test plan, acceptance criteria,
decision/risk registers, and authoritative reference catalog.
- Standard-library documentation/publication validator and focused rejection-case unit tests.
- Future infrastructure/configuration folder guidance and public evidence handling conventions.

## Technical validation and publication

Initial local validation passed for twenty public text candidates, including fifteen Markdown
files: required structure, local links/anchors, twelve stages with all seven required fields,
sixteen requirement mappings, twenty-one test definitions, 100-point rubric, workflow
constraints, and focused publication checks. All twelve unit tests passed using normal filesystem
permissions; the restricted Windows context could not access temporary fixture directories, so
those checks were rerun in the supported context. Independent counts confirmed the proposed 36
GiB guest RAM and 248 GiB guest-disk totals. The completion evidence record adds one reviewed
Markdown file; repository checks are rerun before publishing that record.

The reference catalog contains fifty external URLs. The local HTTPS reachability check confirmed
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HTTP success for thirty-seven; thirteen could not be verified by that client because of
certificate/transport validation issues. Relevant primary-source pages were also researched
through the browsing tool. No TLS verification was disabled. An obsolete Google Help link was
removed rather than published as verified. These external-reference limits do not affect local
Markdown links, and references must be rechecked at stage entry.

| Publication item | Observed result |
| --- | --- |
| Repository |
[waqashamdani-ca/enterprise-zero-trust-security](https://github.com/waqashamdani-ca/enterprise-zer
o-trust-security) |
| Authenticated owner and push actor | `waqashamdani-ca` |
| Permissions | Administrator and push permissions confirmed by CLI/API and connected GitHub
metadata |
| Visibility / default branch | PUBLIC / `main` |
| Origin | `https://github.com/waqashamdani-ca/enterprise-zero-trust-security.git` |
| Initial bootstrap commit |
[2640261ad2738f1e6d4497ee68636621ed403ea3](https://github.com/waqashamdani-ca/enterprise-zero-trus
t-security/commit/2640261ad2738f1e6d4497ee68636621ed403ea3) |
| Initial documentation CI | [Run 42139263675:
SUCCESS](https://github.com/waqashamdani-ca/enterprise-zero-trust-security/actions/runs/4213926367
5), including validation, unit tests, and whitespace steps |
| Synchronization snapshot | Local `main`, `origin/main`, remote HEAD, and remote
`refs/heads/main` matched the initial bootstrap SHA; clean worktree |
| Credentials | Existing secure Git Credential Manager authentication used; no project token or
global Git configuration change |
| Owner UAT / stage execution | PENDING / NOT STARTED |

The initial CLI-backed push was rejected because its OAuth token lacked the workflow scope. The
existing Git Credential Manager login successfully published the same reviewed commit, without
force-push, history alteration, token disclosure, or requesting a new credential. GitHub recorded
the expected owner as the push actor.

See [the bootstrap evidence record](../evidence/BOOTSTRAP-VALIDATION.md) for commands and limits.
This report records the verified initial snapshot; its completion-record commit has its own SHA
and CI run, which are verified separately and reported in the final Task 001 response. Avoid
embedding a commit's own hash inside itself. A passing documentation check is not operational
security verification.

Publication includes only reviewed public-safe text artifacts; actual addressing/inventories,
secrets, raw logs/media, production exports, and proprietary code are excluded. No software
license was added. No Task 001 publication blocker remains; stage-specific design dependencies
and owner UAT remain open.

## Open decisions and risks

Resolve safe egress topology, inspected hardware/disk/backup capacity, exact supported software
versions and evaluation terms, private addressing, lab-only provider identities/plans, log
transports/retention capacity, PKI custody, representative recovery scope, licensing, and
repository protection/reviewer choices. See [DECISIONS.md](DECISIONS.md) and
[RISKS.md](RISKS.md). None is silently approved by publication.

## Recommended next single task and stop

After owner review, obtain explicit approval for **Stage A: read-only hardware and isolation
readiness assessment**. Stop after Task 001. No Proxmox installation, VM creation, OPNsense
configuration, AD deployment, remote-access enrollment, or security validation is authorized by
this report.

## Future final implementation report requirements

This section is a reporting contract for later stages, not completed project results. Replace it
with actual, reviewed outcomes only after the implementation scope is approved and verified.

| Required report element | Evidence needed before claiming completion |
| --- | --- |
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| Executive/business outcome | Actual synthetic workflows supported, scope, owner UAT decision,
and limitations |
| Architecture as built | Sanitized zone/role/capacity diagrams, versions, approved deviations,
and private inventory reference without location disclosure |
| Requirements traceability | R01-R16 -> test cases -> sanitized evidence IDs ->
verdicts/defects/exceptions |
| Identity and PKI | Actual synthetic role counts, effective policy, trust/revocation/expiry and
privileged-access evidence |
| Remote-access evaluation | Separate pilot conditions, scope/feature matrix, denial/revocation
timings, and owner decision |
| Monitoring | Actual source classes, sample denominator/latencies, clock offsets, retention
calculation and observed interval |
| Automation and recovery | Two bounded operations, repeat-run measurements, backup scope/age,
measured RPO/RTO, and restored service/data checks |
| Security limitations | Open defects, residual risks/owner acceptance, unsupported
MFA/posture/revocation paths, and single-host limits |
| Assessment and portfolio | Evidence-based rubric scores and precise lab-scoped resume
achievements |
| Repository and handover | Reviewed PRs/commits, CI, synchronization, operating/recovery
instructions, next bounded task |

Do not fabricate screenshots, test outputs, observed metrics, independent reviewer participation,
acceptance decisions, or production impact.
```